

EXPANDING KALI LINUX DISK SPACE USING GPARTED IN VIRTUALBOX

PROBLEM:

While working on my Kali Linux virtual machine, I discovered that the root partition had reached 100% disk usage. As a result, I could no longer install applications or save additional files. Although I had already increased the virtual hard disk size in VirtualBox to 60 GB, Kali Linux was still using only 38 GB. I needed to expand the existing partition to use the remaining disk space without deleting my files or reinstalling the operating system.

└─\$ df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	1.4G	0	1.4G	0%	/dev
tmpfs	301M	1.4M	299M	1%	/run
/dev/sda1	38G	38G	0	100%	/
tmpfs	1.5G	4.0K	1.5G	1%	/dev/shm
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	1.5G	340K	1.5G	1%	/tmp
Kalishare	238G	212G	26G	90%	/media/sf_Kalishare
tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	301M	100K	301M	1%	/run/user/1000

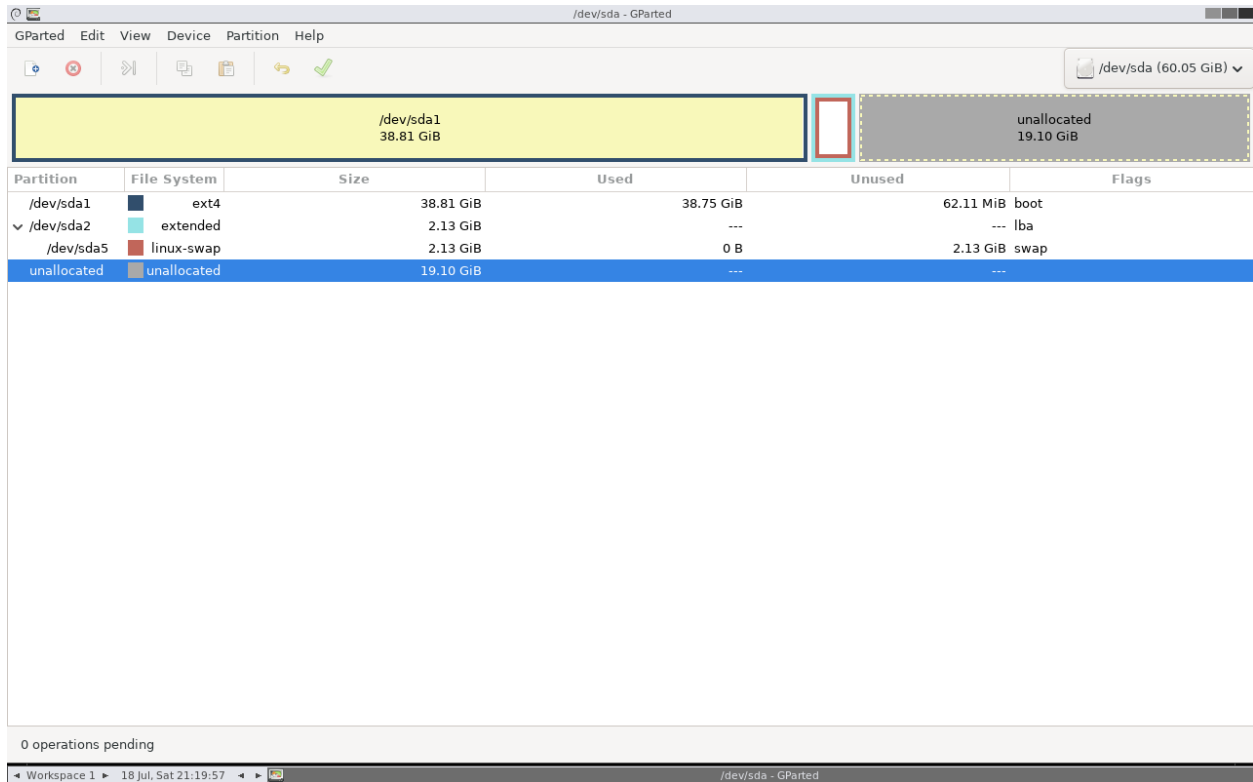
ROOT CAUSE ANALYSIS:

The 19.10 GiB free space was inside an extended partition (/dev/sda2).

Layout was:

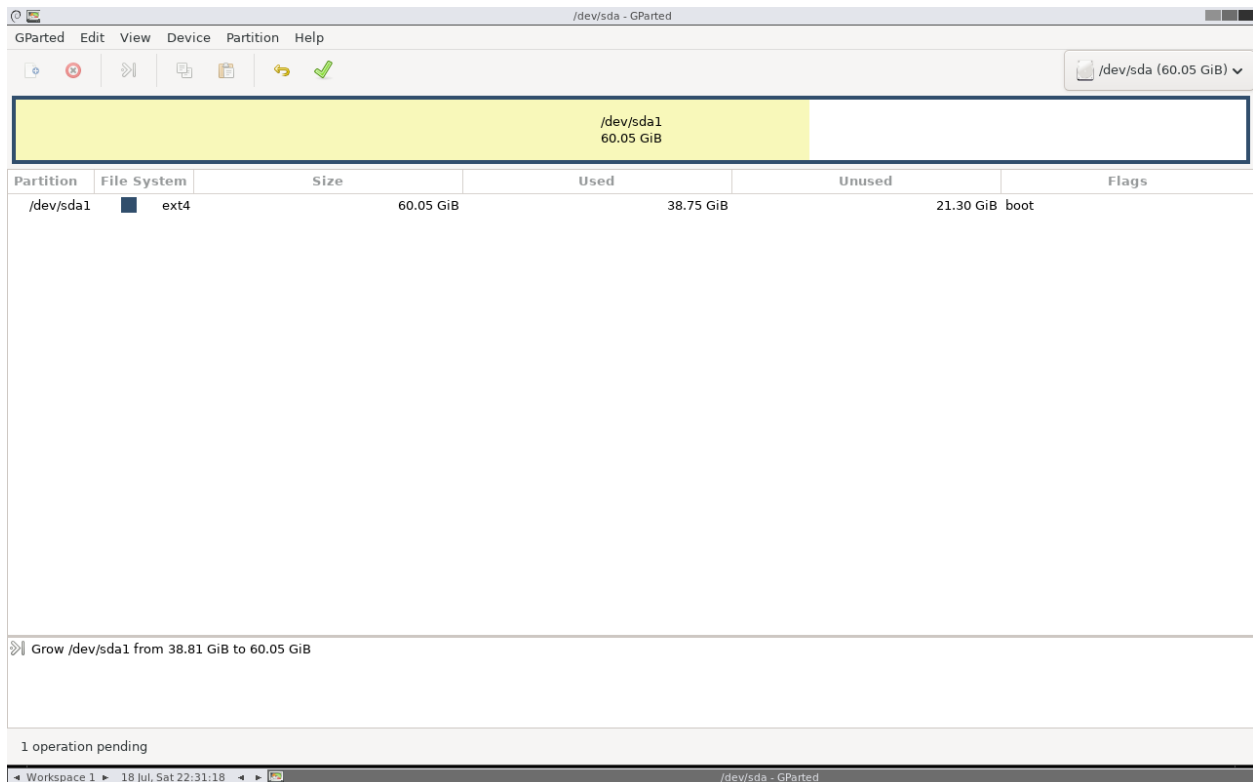
`/dev/sda1 [primary, boot] | /dev/sda2 [extended] -> [unallocated 19GB + /dev/sda5 swap]`

GParted cannot grow a primary partition across the boundary of an extended partition. The free space needed to be moved outside sda2 first.



Solution:

1. Disabled swap: `sudo swapoff -a` and in GParted > Swapoff
2. Deleted logical partition /dev/sda5 (linux-swap 2.13 GiB) - safe, contains no persistent data
3. Deleted extended container /dev/sda2 (21.23 GiB)
4. This resulted in clean layout: `sda1 38.81GB | unallocated 21.24GB`
5. Resized /dev/sda1 from 38.81 GiB to ~60 GiB using Resize/Move
6. Applied all operations and verified with `df -h` and successful reboot



Error Handled:

`GParted Bug: A partition cannot end after the end of the device` - caused by setting Free space preceding without reducing New Size. Fixed by deleting extended container instead of moving left edge.

